

Solution of Problems of M670_7 (Hypothesis 1)

Problem 1: In the past, 75% of the tourists who visited Miami went to see Miami Beach. The management of Miami Beach recently undertook a promotional campaign. They are interested in determining whether the campaign actually increased the proportion of tourists visiting Miami Beach.

1. What is the correct set of hypotheses?

The correct set of hypotheses is:

$$H_0 : p = 0.75$$

$$H_1 : p > 0.75$$

where p is the proportion of tourists visiting Miami Beach.

2. Given that the p-value is 0.03, what is your conclusion?

Since the p-value (0.03) is less than the common significance level (0.05), we reject the null hypothesis. This suggests that there is sufficient evidence to support the claim that the promotional campaign increased the proportion of tourists visiting Miami Beach.

3. Assume that in reality the promotional campaign actually increased the proportion of tourists visiting Miami Beach. Did your decision in part b make any error? If yes, what type of error? What if p-value=0.3?

If the decision was to reject the null hypothesis when the p-value was 0.03, and the campaign did indeed increase the proportion, there was no error.

However, if the p-value were 0.3, the decision would be to fail to reject the null hypothesis. In this case, since the campaign did increase the proportion, failing to reject the null hypothesis would result in a Type II error.

Problem 2: Your investment executive claims that the average yearly rate of return on the stocks she recommends is at least 10.0%. You plan on taking a sample to test her claim.

1. What is the correct set of hypotheses?

The correct set of hypotheses is:

$$H_0 : \mu \geq 10.0\%$$

$$H_1 : \mu < 10.0\%$$

where μ is the average yearly rate of return on the stocks.

2. Given that the p-value is 0.03, what is your conclusion?

Since the p-value (0.03) is less than the common significance level (0.05), we reject the null hypothesis. This suggests that there is sufficient evidence to support the claim that the average yearly rate of return on the stocks is less than 10.0%.

3. Assume that in reality, the average yearly rate of return on the stocks she recommends is indeed at least 10.0%. Did your decision in part b make any error? If yes, what type of error? What if p-value=0.3?

If the decision was to reject the null hypothesis when the p-value was 0.03, and the actual average yearly rate of return is indeed at least 10.0%, then rejecting the null hypothesis would result in a Type I error.

If the p-value were 0.3, the decision would be to fail to reject the null hypothesis. In this case, since the average rate of return is indeed at least 10.0%, failing to reject the null hypothesis would not result in an error.

Problem 3: A major west coast city provides one of the most comprehensive emergency medical services in the world. Operating with approximately 20 mobile medical units, the city has maintained the service goal to respond to medical emergencies with a mean time of 12 minutes or less. The manager of this service wanted to test if the 12 minutes standard still hold. A sample of 40 response times was collected, and the average response time was 13.25 minutes. Population standard deviation is assumed to be 3.2 minutes. What can be concluded regarding the average response time of the city's medical emergencies? with 0.05 level of confidence? with 0.01 level of confidence?

Hypotheses: Denote μ as the average response time.

$$H_0 : \mu \leq 12$$

$$H_1 : \mu > 12$$

Data: - Sample mean ($\bar{x}$): 13.25 minutes - Population standard deviation (σ): 3.2 minutes - Sample size (n): 40

Test Statistic:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{13.25 - 12}{3.2/\sqrt{40}} \approx 2.47$$

P-value: = 0.0068.

At the 0.05 or 0.01 significance level: Since the p-value (0.0136) is less than 0.05 and 0.01, we reject the null hypothesis. There is sufficient evidence to conclude that the average response time is greater than 12 minutes.

Option 2: Use Critical Values: - For $\alpha = 0.05$: $z_{0.05} \approx 1.645$ - For $\alpha = 0.01$: $z_{0.01} \approx 2.33$

At the 0.05 significance level: Since the test statistic $z = 2.47$ is greater than the critical value 1.645, we reject the null hypothesis. This indicates that the average response time is likely greater than 12 minutes at the 5% significance level.

At the 0.01 significance level: Since the test statistic $z = 2.47$ is greater than the critical value 2.33, we reject the null hypothesis. This indicates that the average response time is also likely greater than 12 minutes at the 1% significance level.

Problem 4: A major west coast city provides one of the most comprehensive emergency medical services in the world. Operating in a multiple hospital system with approximately 20 mobile medical units, the city has maintained the service goal to respond to medical emergencies with a mean time of 12 minutes or less. The manager of this service wanted to test if the 12 minutes standard still hold. A sample of 40 response times was collected, sample mean = 12.535, sample standard deviation = 4.18. What can be concluded regarding the average response time of the city's medical emergencies? with 0.05 level of confidence? with 0.01 level of confidence?

Hypothesis:

- Null Hypothesis (H_0): $\mu \leq 12$ minutes
- Alternative Hypothesis (H_1): $\mu > 12$ minutes

Test Statistic:

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

where:

- $\bar{x}$ is the sample mean (12.535 minutes).
- μ_0 is the hypothesized mean under the null hypothesis (12 minutes).
- s is the sample standard deviation (4.18 minutes).
- n is the sample size (40).

Given:

- Sample mean ($\bar{x}$) = 12.535 minutes
- Hypothesized mean (μ_0) = 12 minutes
- Sample standard deviation (s) = 4.18 minutes
- Sample size (n) = 40

Calculate:

$$\frac{s}{\sqrt{n}} = \frac{4.18}{\sqrt{40}} \approx 0.662$$

$$t = \frac{12.535 - 12}{0.662} \approx \frac{0.535}{0.662} \approx 0.81$$

Degrees of Freedom:

$$df = n - 1 = 40 - 1 = 39$$

Decision Rule:

- At the 0.05 significance level (one-tailed test), the critical value is approximately 1.685. Since $t = 0.81 < 1.685$, do not reject H_0 .
- At the 0.01 significance level (one-tailed test), the critical value is approximately 2.423. Since $t = 0.81 < 2.423$, do not reject H_0 .

Conclusion:

- At the 0.05 significance level, we do not reject the null hypothesis and conclude that there is not enough evidence to suggest that the average response time is greater than 12 minutes.
- At the 0.01 significance level, we do not reject the null hypothesis and conclude that there is not enough evidence to suggest that the average response time is greater than 12 minutes.

Problem 5: The production line for Glow toothpaste is designed to fill tubes of toothpaste with a target mean weight of 6 ounces. Periodically, a sample of 30 tubes will be selected in order to check the filling process. Quality assurance procedures call for the continuation of the filling process if the sample results are consistent with the assumption that the mean filling weight for the population of toothpaste tubes is 6 ounces; otherwise the filling process will be stopped and adjusted. One day, the sample mean is found to be 6.1 ounces; sample standard deviation is .2 ounces. The floor manager does not want to stop the process and argue that the .1 ounce difference between the sample mean and target is “not a big deal.” What do you think?

Hypothesis:

- Null Hypothesis (H_0): $\mu = 6$ ounces
- Alternative Hypothesis (H_1): $\mu \neq 6$ ounces

Test Statistic:

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

where:

- $\bar{x}$ is the sample mean (6.1 ounces).
- μ_0 is the hypothesized population mean (6 ounces).
- s is the sample standard deviation (0.2 ounces).
- n is the sample size (30).

Given:

- Sample mean ($\bar{x}$) = 6.1 ounces
- Hypothesized mean (μ_0) = 6 ounces
- Sample standard deviation (s) = 0.2 ounces
- Sample size (n) = 30

Calculate:

$$\frac{s}{\sqrt{n}} = \frac{0.2}{\sqrt{30}} \approx 0.0365$$
$$t = \frac{6.1 - 6}{0.0365} \approx \frac{0.1}{0.0365} \approx 2.74$$

Degrees of Freedom:

$$df = n - 1 = 30 - 1 = 29$$

Decision Rule:

- At the 0.05 significance level (two-tailed test), the critical value is approximately 2.045. Since $|t| = 2.74 > 2.045$, reject H_0 .
- At the 0.01 significance level (two-tailed test), the critical value is approximately 2.756. Since $|t| = 2.74 < 2.756$, do not reject H_0 .

Conclusion:

- At the 0.05 significance level, we reject the null hypothesis and conclude that there is significant evidence that the mean filling weight is different from 6 ounces. Therefore, it may be prudent to adjust the filling process.
- At the 0.01 significance level, we do not reject the null hypothesis, suggesting less strong evidence of a deviation from the target mean. However, a review of the process is still advisable.

Problem 6: You're an accounting manager. A year-end audit showed 6% of transactions had errors. You implement new procedures. A random sample of 500 transactions had 25 errors. Has the proportion of incorrect transactions decreased at the .05 level?

To test if the proportion of incorrect transactions has decreased, we perform a one-sample z -test for proportions.

Hypothesis:

- Null Hypothesis (H_0): $p \geq 0.06$
- Alternative Hypothesis (H_1): $p < 0.06$

Test Statistic:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

where:

- $\hat{p}$ is the sample proportion of errors.
- p_0 is the hypothesized population proportion (0.06).
- n is the sample size.

Given:

- Number of errors in the sample = 25
- Sample size (n) = 500
- Hypothesized proportion (p_0) = 0.06

Calculate:

$$\hat{p} = \frac{25}{500} = 0.05$$

$$\text{Standard Error} = \sqrt{\frac{p_0(1-p_0)}{n}} = \sqrt{\frac{0.06 \times (1-0.06)}{500}} = \sqrt{\frac{0.0564}{500}} \approx 0.0106$$

$$z = \frac{\hat{p} - p_0}{\text{Standard Error}} = \frac{0.05 - 0.06}{0.0106} \approx -0.945$$

Decision Rule: For a one-tailed test at the 0.05 significance level, the critical value is approximately -1.645.

- Since $z = -0.945$ is greater than -1.645, we do not reject H_0 .

Conclusion: There is not enough statistical evidence at the 0.05 level to conclude that the proportion of incorrect transactions has decreased.

EX 1: The manager of a grocery store has taken a random sample of 100 customers. The average length of time it took the customers in the sample to check out was 3.1 minutes with a standard deviation of 0.5 minutes. We want to test to determine whether or not the mean waiting time of all customers is significantly more than 3 minutes. Test with 95% confidence.

1. Formulate the Hypotheses

- Null Hypothesis (H_0): The mean waiting time is 3 minutes.

$$H_0 : \mu = 3$$

- Alternative (H_1): The mean waiting time is more than 3 minutes.

$$H_1 : \mu > 3$$

2. Calculate the Test Statistic

Use the sample data to calculate the test statistic t :

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

where:

- $\bar{x} = 3.1$ (sample mean),
- $\mu_0 = 3$ (hypothesized population mean),
- $s = 0.5$ (sample standard deviation),
- $n = 100$ (sample size).

Substitute the values: $t = \frac{3.1-3}{\frac{0.5}{\sqrt{100}}} = 2$.

3. Determine the Critical Value and Make a Decision

Degrees of freedom = $n - 1 = 99$. For a one-tailed test with $\alpha = 0.05$, the critical value for the t-test is approximately 1.66. In this case:

$$t = 2 > 1.66$$

we reject the null hypothesis. At the 95% confidence level, there is sufficient evidence to conclude that the mean waiting time of all customers is significantly more than 3 minutes.

EX 2: A lathe is set to cut bars of steel into lengths of 6 centimeters. The lathe is considered to be in perfect adjustment if the average length of the bars it cuts is 6 centimeters. A sample of 121 bars is selected randomly and measured. It is determined that the average length of the bars in the sample is 6.08 centimeters with a standard deviation of 0.44 centimeters. Test this with 95% confidence.

1. Formulate the Hypotheses

- Null Hypothesis (H_0): The average length of the bars is 6 centimeters.

$$H_0 : \mu = 6$$

- Alternative Hypothesis (H_1): The average length of the bars is not 6 centimeters.

$$H_1 : \mu \neq 6$$

2. Set the Significance Level

For a two-tailed test with $\alpha = 0.05$, the critical values for the z-test are ± 1.96 . In this problem, $n = 121$ is large, so z approximation is fine. We can also use t-test statistic. The conclusion will be similar.

3. Calculate the Test Statistic

Given:

- Sample size, $n = 121$,
- Sample mean, $\bar{x} = 6.08$ centimeters,
- Sample standard deviation, $s = 0.44$ centimeters,
- Hypothesized mean, $\mu_0 = 6$ centimeters.

The test statistic z is given by:

$$z = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

Substitute the values:

$$z = \frac{6.08 - 6}{\frac{0.44}{\sqrt{121}}}$$

$$z = \frac{0.08}{\frac{0.44}{11}}$$

$$z = \frac{0.08}{0.04}$$

$$z = 2$$

4. Determine the Critical Values and Make a Decision

For a two-tailed test with $\alpha = 0.05$, the critical values are ± 1.96 .

- If $|z| > 1.96$, reject H_0 .
- If $|z| \leq 1.96$, fail to reject H_0 .

In this case:

$$|z| = 2$$

Since $2 > 1.96$, we reject the null hypothesis.

5. Conclusion

At the 95% confidence level, there is sufficient evidence to conclude that the average length of the bars is significantly different from 6 centimeters.

EX 3: An official of a large national union claims that the fraction of women in the union is not significantly different from one-half. Using the critical value approach and the sample information reported below, carry out a test of this statement. Let $\alpha = 0.05$.

sample size: 400, women: 168; men: 232.

1. Formulate the Hypotheses

- Null Hypothesis (H_0): The fraction of women in the union is 0.5.

$$H_0 : p = 0.5$$

- Alternative Hypothesis (H_1): The fraction of women in the union is not 0.5.

$$H_1 : p \neq 0.5$$

2. Set the Significance Level

For a two-tailed test with $\alpha = 0.05$, the critical value for the z-test is approximately ± 1.96 .

3. Calculate the Test Statistic

Given:

- Sample size, $n = 400$,
- Number of women, $X = 168$,
- Sample proportion, $\hat{p} = \frac{X}{n} = \frac{168}{400} = 0.42$,
- Hypothesized proportion, $p_0 = 0.5$.

The test statistic z is given by:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Substitute the values:

$$z = \frac{0.42 - 0.5}{\sqrt{\frac{0.5 \times (1-0.5)}{400}}}$$

$$\begin{aligned}
 z &= \frac{-0.08}{\sqrt{\frac{0.25}{400}}} \\
 z &= \frac{-0.08}{\sqrt{0.000625}} \\
 z &= \frac{-0.08}{0.025} \\
 z &= -3.2
 \end{aligned}$$

4. Determine the Critical Values and Make a Decision

For a two-tailed test with $\alpha = 0.05$, the critical values are ± 1.96 .

- If $|z| > 1.96$, reject H_0 .
- If $|z| \leq 1.96$, fail to reject H_0 .

In this case:

$$|z| = 3.2$$

Since $3.2 > 1.96$, we reject the null hypothesis.

5. Conclusion

At the 0.05 significance level, there is sufficient evidence to conclude that the fraction of women in the union is significantly different from one-half.